

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Specialist techniques & statistical analysis - Set 1

Mark scheme

AQA spec 3.7.2.3; Research methods; textbook pp. 402-407, 420-423

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	2 marks per point 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. 2 Image databases: identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features 2 eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance – contamination / DNA transport by water 2 Geolocator tags: tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate – animal must be recaptured to download data Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	6

Question	Marking guidance	Marks
01.2	1 mark each 1 mark for each of the 2 points below. 1 Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) 1 Microwaves pass through cloud so monitoring is not interrupted by weather Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.3	Any 2 of 3 1 mark each for any 2 of the 3 points below, in any order, maximum 2. 1 Droppings: diet, gender, territory size 1 Owl pellets: diet 1 Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	
02.1	1 mark each 1 mark for each of the 2 points below. 1 The scatter / variability of results around each mean – about 68% of values lie within one SD of the mean 1 The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. standard deviation → a measure of the spread of the data about the mean	2
02.2	1 mark each 1 mark for each of the 2 points below. 1 t-test 1 Comparing two means of measured (continuous) data Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round. t-test → a test for a difference between the means of two normally distributed sets Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. 'the result was significant' without comparing the calculated value with the critical value at $p = 0.05$	2

Question	Marking guidance	Marks
02.3	1 mark each 1 mark for each of the 2 points below. 1 There is a 30% probability that the difference occurred by random chance 1 Above the 0.05 (5%) level so the difference is not significant – the null hypothesis (no difference) is accepted Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. random → positions chosen using random numbers so the choice is not biased null hypothesis → a statement that there is no significant difference or correlation Do not credit 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
02.4	1 mark each 1 mark for each of the 2 points below. 1 Two correlated variables may both be caused by a third variable that was not investigated 1 Significance only shows the result is unlikely to be chance – a mechanism / controlled experiment is needed for causation Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. statistically significant → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance correlation → a relationship in which the two variables change together causal → one variable actually causing the change in the other Do not credit - a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
02.5	1 mark each 1 mark for each of the 2 points below. 1 Increase the number of replicates (reduces the effect of variability) 1 Standardise other variables (soil, water, light) more closely to reduce scatter Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
03.1	1 mark each 1 mark for each of the 4 points below. 1 Spearman's rank correlation coefficient 1 Chi-squared test (frequencies / counts in categories) 1 t-test (two means of measured data) 1 Spearman's rank correlation coefficient Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Spearman → a test for correlation between two sets of ranked data correlation → a relationship in which the two variables change together Chi-squared → a test comparing observed frequencies with expected frequencies Do not credit - a correlation presented as proof of cause - say that it suggests a link that further work would have to test 'the result was significant' without comparing the calculated value with the critical value at $p = 0.05$	4
03.2	1 mark for the point below. 1 Chi-squared compares frequencies / counts in categories – it cannot be used for continuously variable measured data Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. chi-squared → a test comparing observed frequencies with expected frequencies Do not credit 'the result was significant' without comparing the calculated value with the critical value at $p = 0.05$	1
03.3	1 mark each 1 mark for each of the 3 points below. 1 The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level 1 The negative sign shows a negative correlation: one variable decreases as the other increases 1 There is less than a 5% probability the correlation arose by chance – reject the null hypothesis (but correlation is not proof of causation) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Spearman → a test for correlation between two sets of ranked data critical value → the table value the calculated statistic must reach to be significant $p = 0.05$ → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance Do not credit - a correlation presented as proof of cause - say that it suggests a link that further work would have to test 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	3

Question	Marking guidance	Marks
03.4	1 mark each 1 mark for each of the 2 points below. 1 Show the standard deviation / range of the original values around each mean 1 If the bars of two means overlap greatly the difference is probably not significant – little overlap suggests a significant difference Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. standard deviation → a measure of the spread of the data about the mean	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
04.1	1 mark each 1 mark for each of the 3 points below. 1 Each regional population has its own unique gene pool / DNA profile 1 DNA from seized timber or ivory is compared with the database of regional populations 1 This identifies where the material came from, showing whether it was legally harvested and where poaching occurs Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. gene pool → the range of alleles / genes present in a population	3
04.2	1 mark each 1 mark for each of the 2 points below. 1 Radio / GPS transmitters use radio waves - work in air and fresh water, giving positions (GPS stores or transmits location) 1 Radio waves do not travel through seawater so acoustic transmitters (sound) are used for marine animals, giving real-time positions Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round.	2
04.3	Marks shared Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. up to 2 Drones / ROVs, AUVs (eg Autosub), balloons, aircraft, satellites, animals (tags) up to 2 Drones: cheap, low disturbance, high-resolution images of areas hard to reach (cliffs, canopy, ice), repeated surveys – infrared cameras detect animals at night Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. infrared → heat radiation / long-wave radiation re-emitted by the Earth Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Question	Marking guidance	Marks
04.4	1 mark each 1 mark for each of the 2 points below. 1 Cameras record automatically (visible light by day, infrared at night) without researchers being present 1 So animals behave naturally and nests are not deserted – monitoring continues 24 hours Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. infrared → heat radiation / long-wave radiation re-emitted by the Earth	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-406; AQA spec 3.7.2.3.	
05.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 $\Sigma n(n - 1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ 1 $N(N - 1) = 100 \times 99 = 9900$ 1 $D = 9900 / 2900 = 3.41$ (accept 3.4) Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. Simpson → a diversity index that uses both the number of species and their abundance Do not credit - A bare wrong answer with no working. 'more species always means a higher index' - the evenness of the abundances matters as well	3
05.2	1 mark each 1 mark for each of the 2 points below. 1 Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B 1 The index does not identify which species are present / a rare or important species may be missed – depends on identification skill and sampling method Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round. biodiversity → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
06.1	1 mark each 1 mark for each of the 3 points below. 1 t-test 1 The data are measured (continuous) values and two means are being compared 1 The data are (assumed to be) normally distributed Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round. t-test → a test for a difference between the means of two normally distributed sets Do not credit 'the result was significant' without comparing the calculated value with the critical value at $p = 0.05$	3
06.2	1 mark for the point below. 1 Use the Mann-Whitney U test instead (compares medians) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Mann-Whitney → a test for a difference between two sets of unpaired, non-normal data Do not credit 'the result was significant' without comparing the calculated value with the critical value at $p = 0.05$	1
06.3	1 mark each 1 mark for each of the 2 points below. 1 There is a 2% probability that the difference occurred by random chance 1 As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. random → positions chosen using random numbers so the choice is not biased statistically significant → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance null hypothesis → a statement that there is no significant difference or correlation Do not credit 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
06.4	1 mark each 1 mark for each of the 2 points below. 1 The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 1 About 68% of values lie within one SD of each mean so the two data sets are clearly separated Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
06.5	1 mark each 1 mark for each of the 2 points below. 1 Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists 1 Another variable (eg topography / aspect of the sites) could cause both – a controlled experiment or further evidence is needed Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. statistically significant → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance causal → one variable actually causing the change in the other	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
07.1	1 mark each 1 mark for each of the 4 points below. 1 Spearman's rank correlation coefficient 1 Mann-Whitney U test 1 Chi-squared test 1 t-test Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Spearman → a test for correlation between two sets of ranked data correlation → a relationship in which the two variables change together Mann-Whitney → a test for a difference between two sets of unpaired, non-normal data Do not credit - a correlation presented as proof of cause - say that it suggests a link that further work would have to test 'the result was significant' without comparing the calculated value with the critical value at $p = 0.05$	4
07.2	1 mark each 1 mark for each of the 2 points below. 1 Chi-squared compares frequencies / counts in categories 1 Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round. chi-squared → a test comparing observed frequencies with expected frequencies t-test → a test for a difference between the means of two normally distributed sets Mann-Whitney → a test for a difference between two sets of unpaired, non-normal data Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. 'the result was significant' without comparing the calculated value with the critical value at $p = 0.05$	2

Question	Marking guidance	Marks
07.3	1 mark each 1 mark for each of the 2 points below. 1 There is always some remaining uncertainty / a probability that the result was produced by chance 1 Statistics only give a level of confidence (eg 95%, 99%) that the result is significant Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	1 mark each 1 mark for each of the 2 points below. 1 Bars drawn above and below each mean showing plus and minus one standard deviation 1 If the bars of two means overlap a lot the difference is probably not significant – if they do not overlap it probably is Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. standard deviation → a measure of the spread of the data about the mean	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	